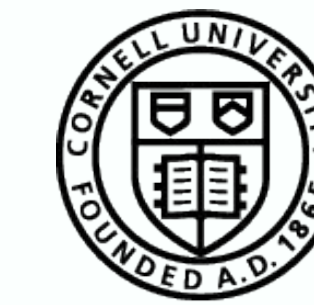


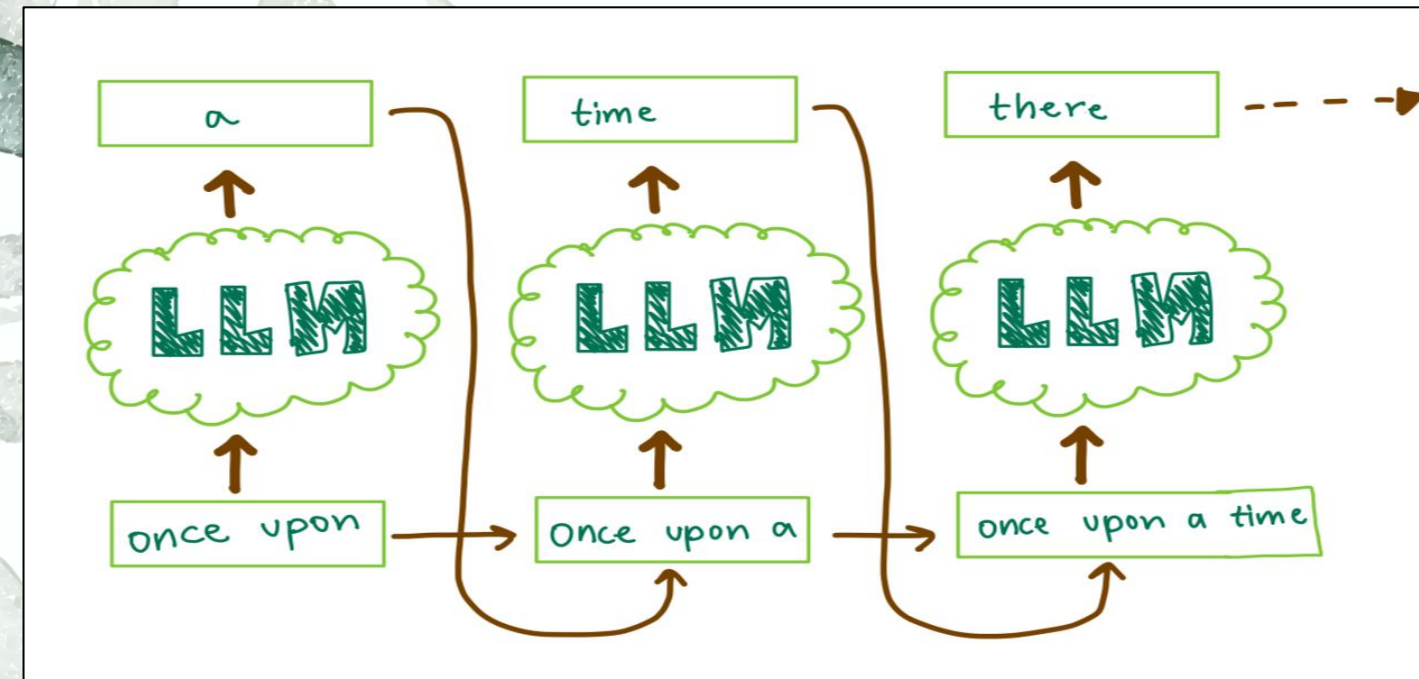
Tree of Thoughts



CornellBowers
Computer Science

Aarsha Joshi (aj569), Aileen Huang (aeh245), Bella Falkenberg (alf253), Ena Kovac (ehk75)

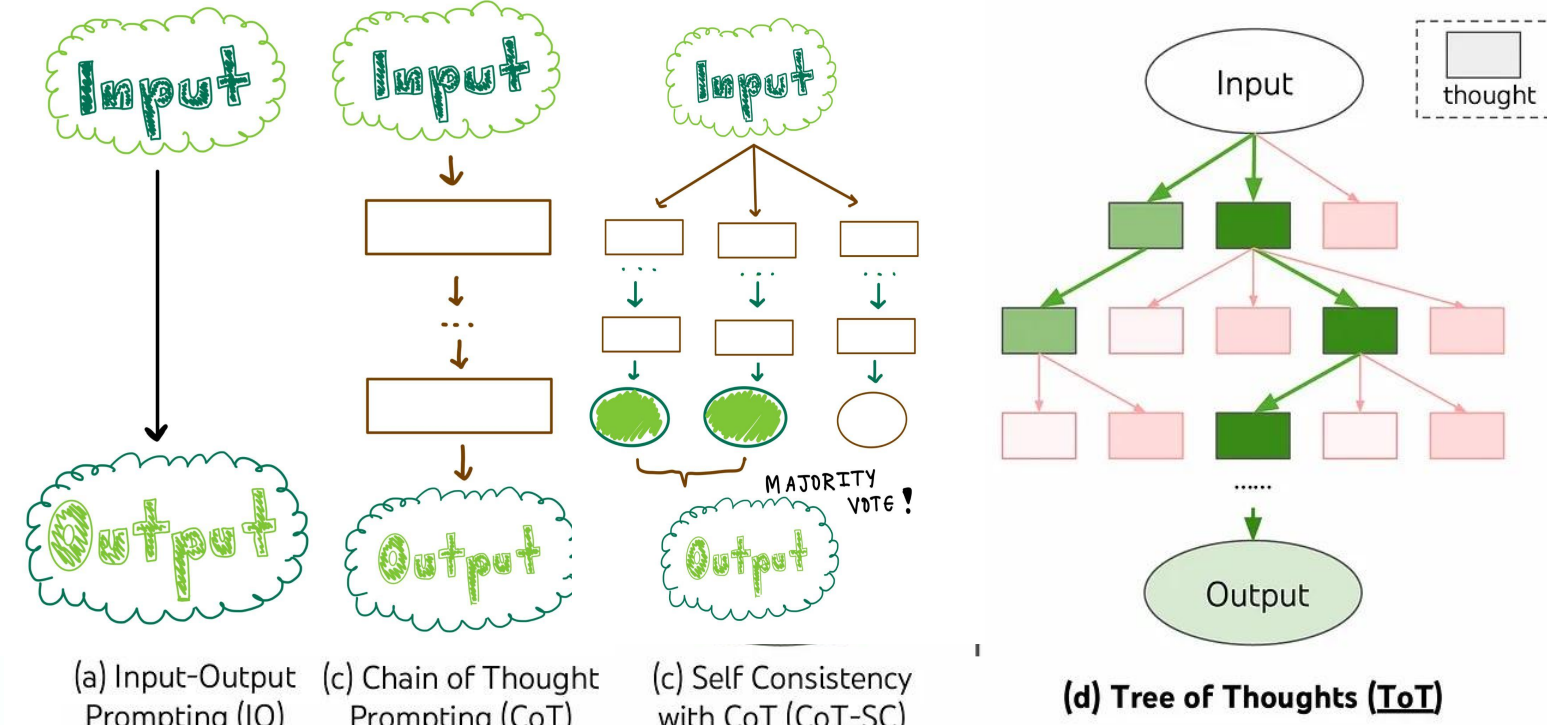
MOTIVATION



Traditional LLMs generate responses left-to-right making one decision at a time.

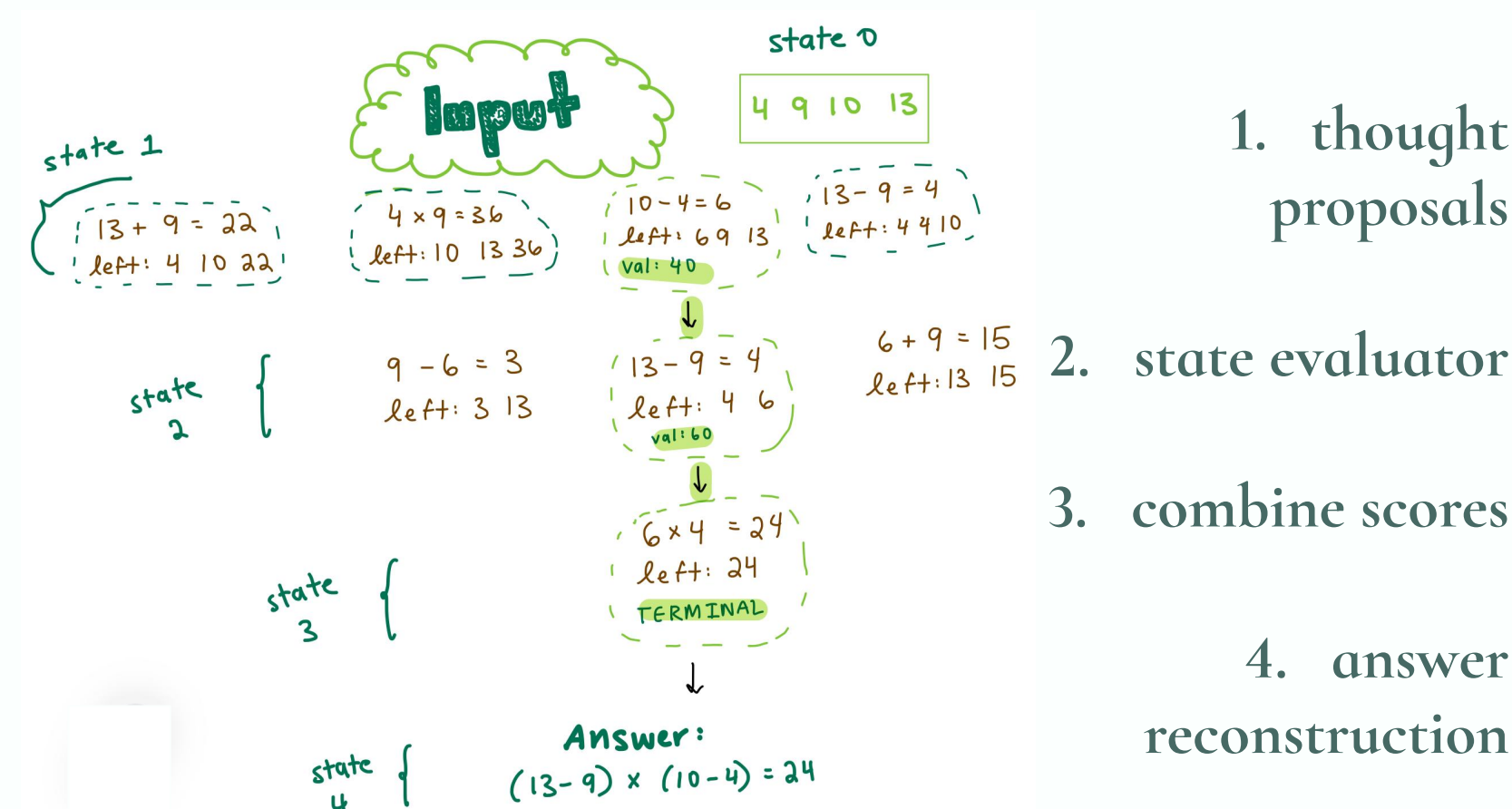
Tree of Thoughts (ToT): allows LLMs to explore multiple reasoning paths

METHODOLOGY



Organizes reasoning steps as nodes in a search tree and uses search algorithms to explore paths.

Game of 24



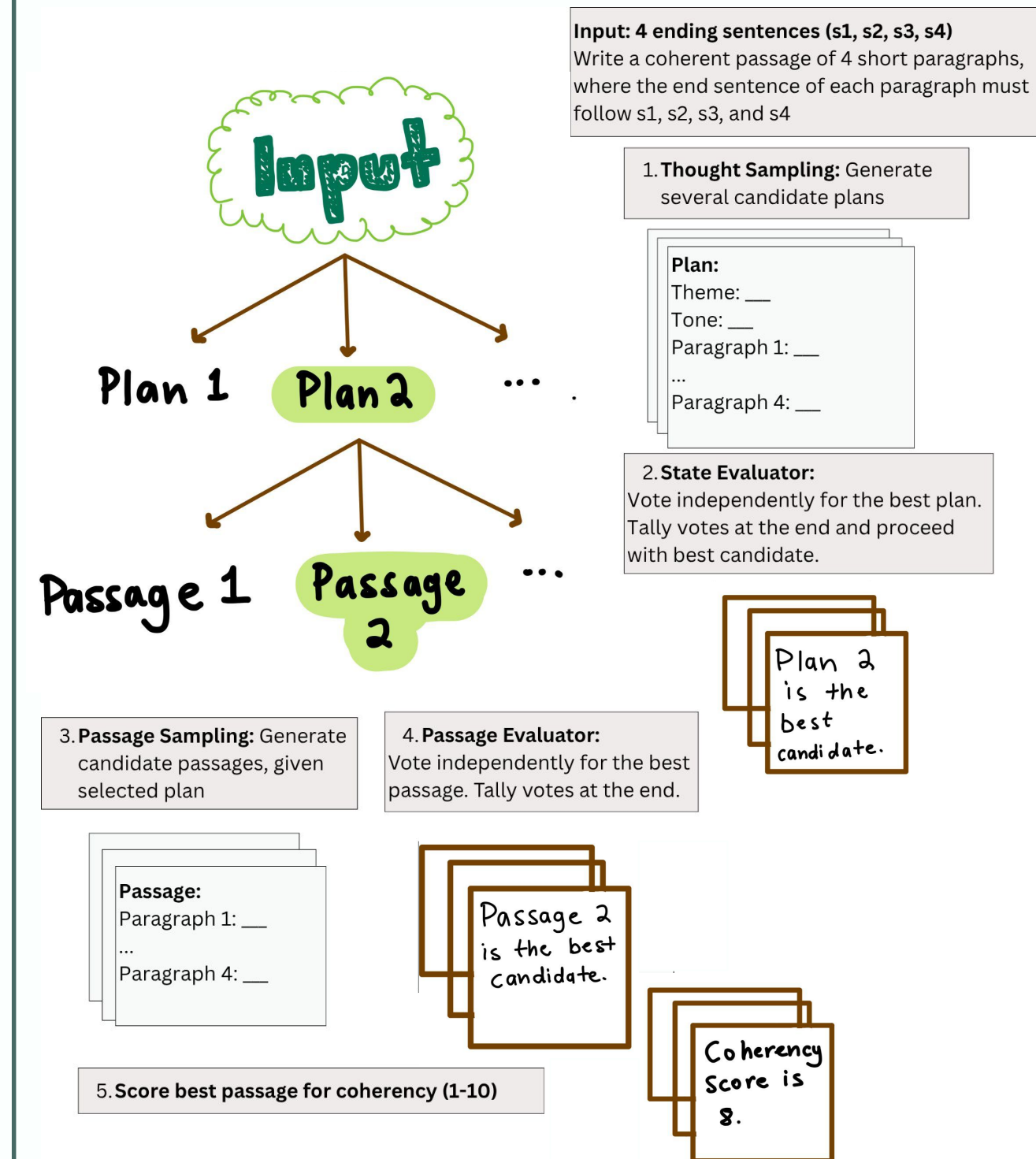
1. thought proposals

2. state evaluator

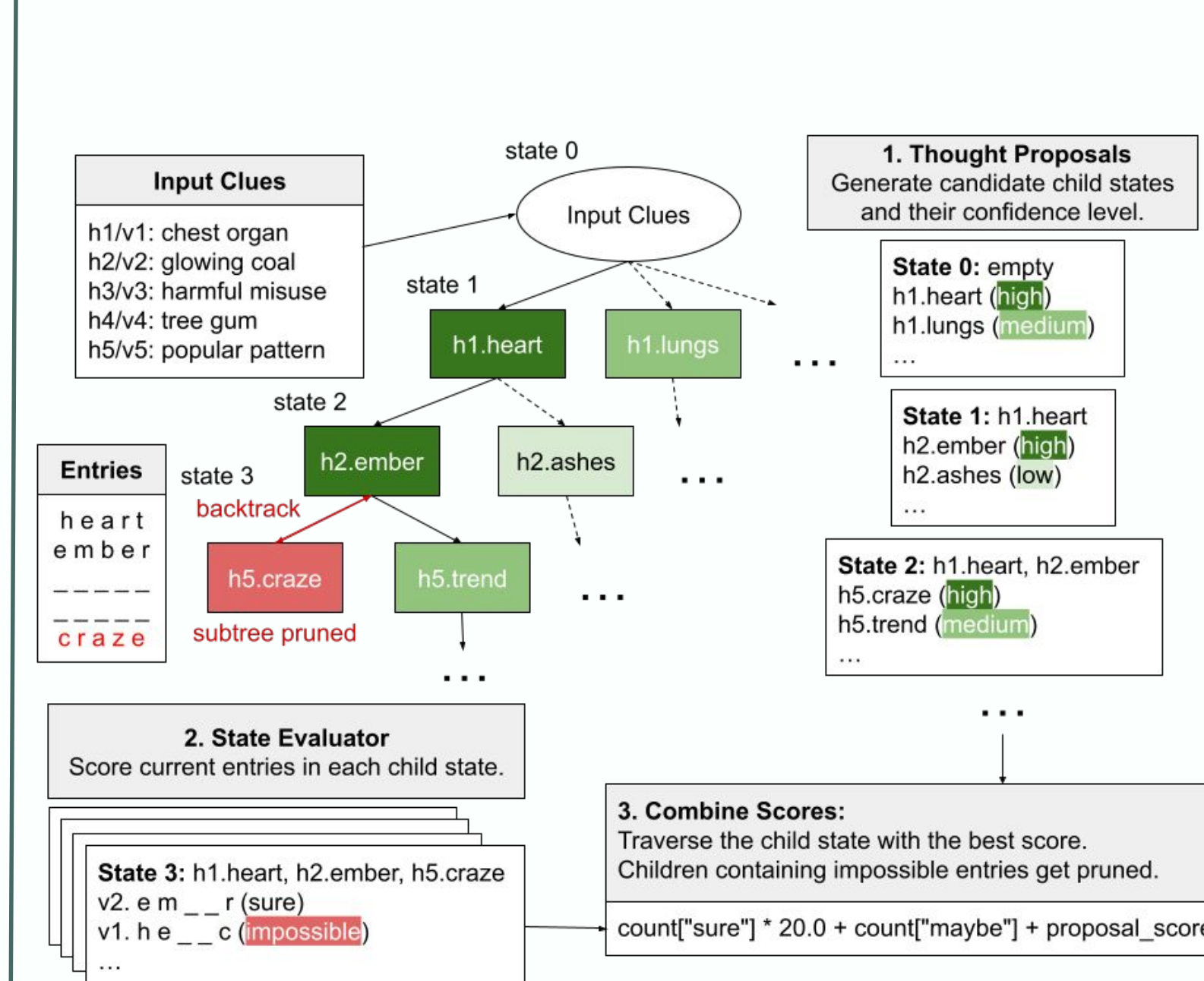
3. combine scores

4. answer reconstruction

Creative Writing

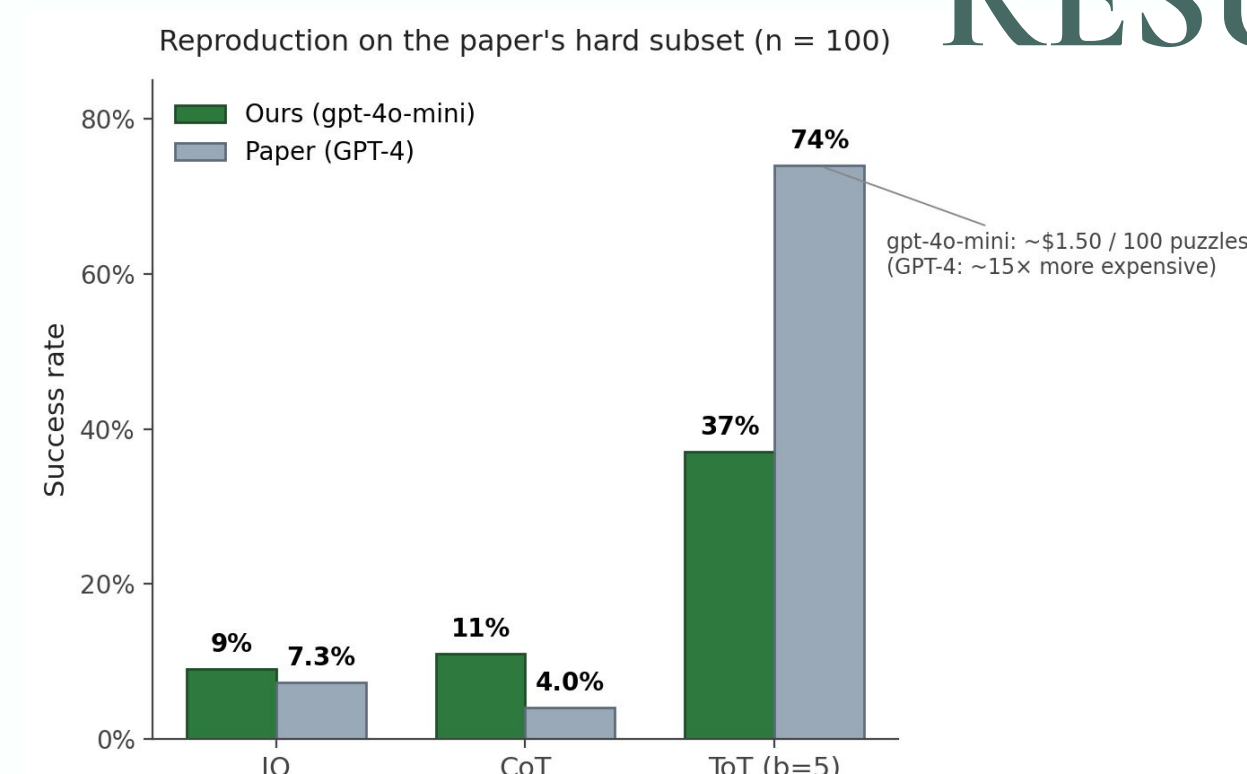


Mini Crosswords

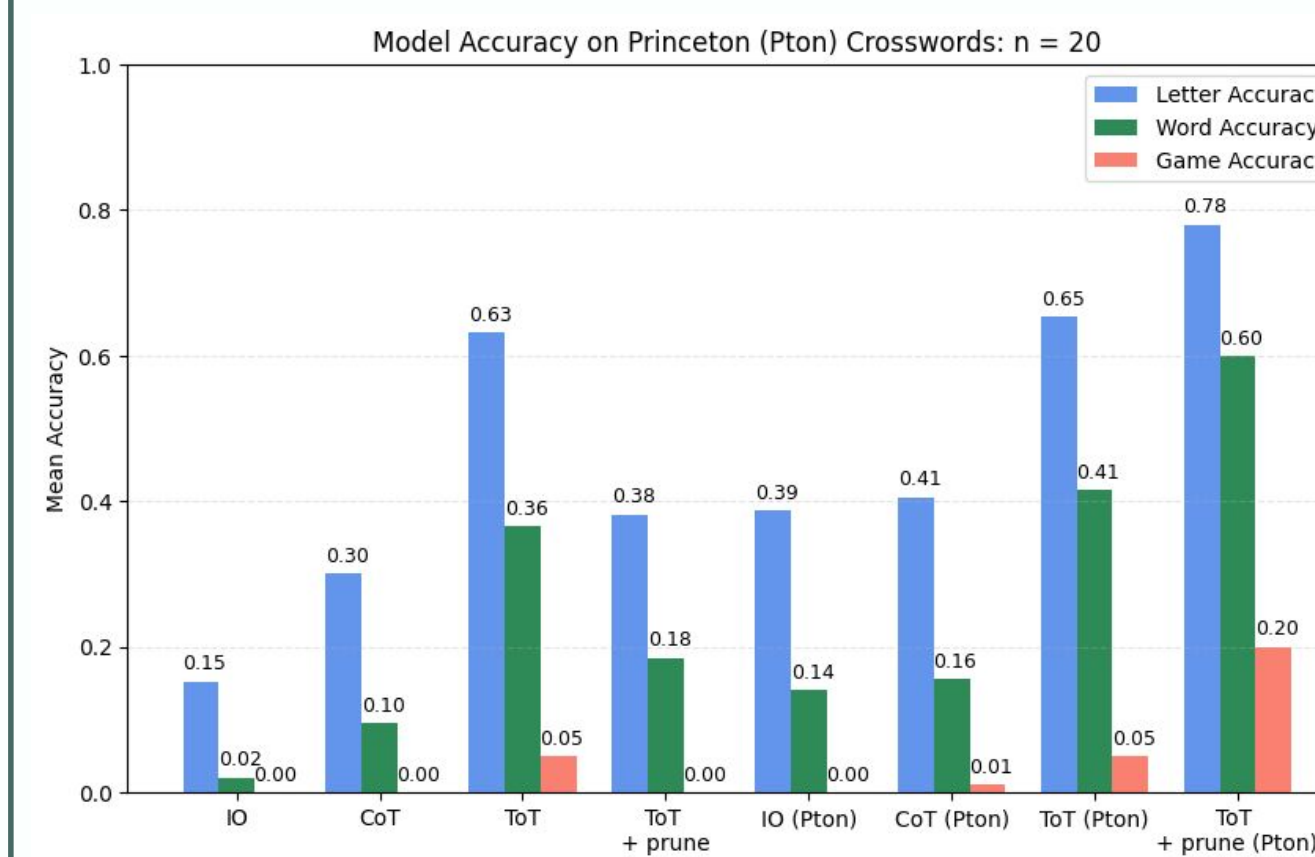


RESULTS

Game of 24

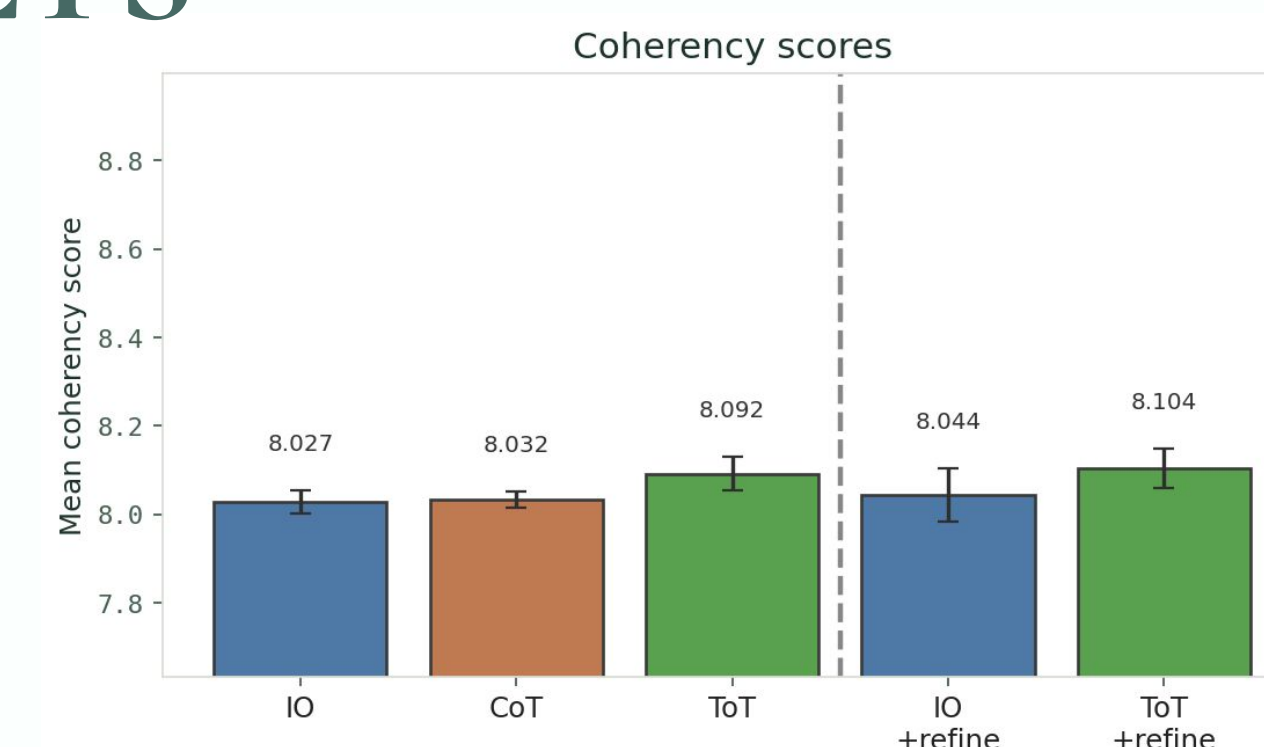


Mini Crosswords



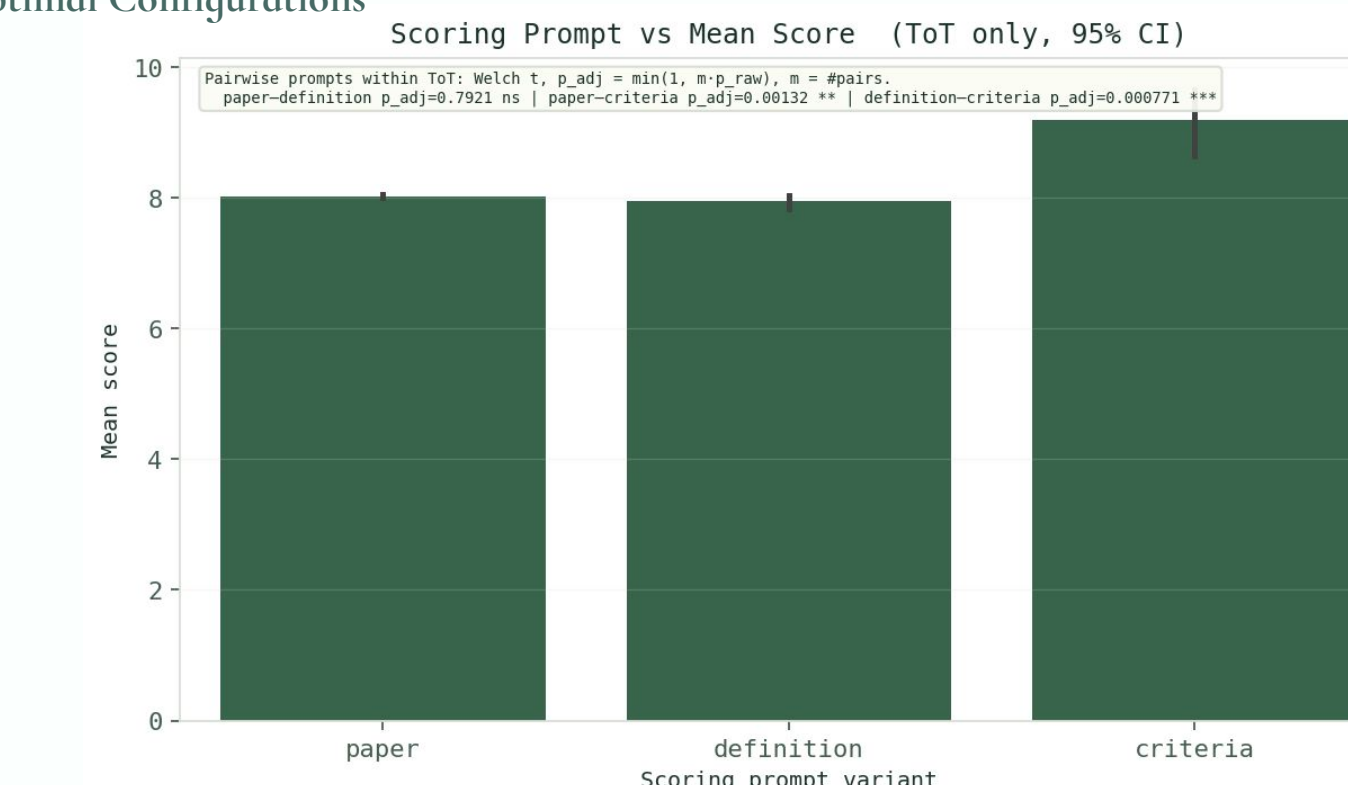
- ToT consistently performed better than IO and CoT
- Pruning worsened performance; discouraged obscure words

Creative Writing



Replicated ToT-refine > ToT > CoT > IO.
ToT preferred 14/20 pairwise (41/100 in the paper)

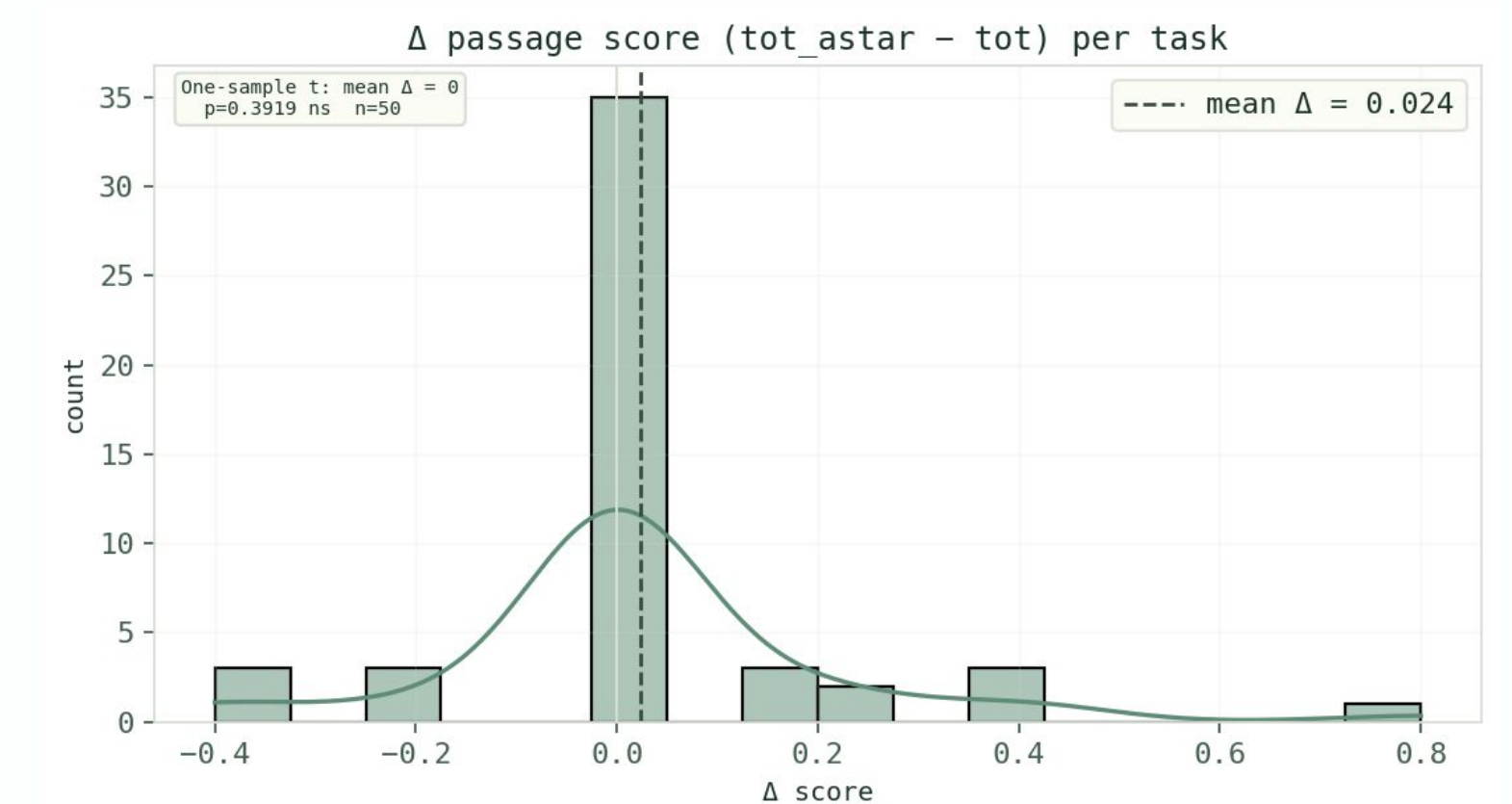
Optimal Configurations



Optimal: Criteria-based scoring and GPT-4.1-mini scorer with ToT-refine to optimize score, and k=5 candidates with n≥5 votes for maximum vote diversity.

EXTENDED WORK

- Using a Weaker Model
- Hallucination Reduction
 - Identifying recognized vs unrecognized patterns
- Creative Writing - strategies to Improve Planning Step



Tasks where A* scored higher than ToT also tended to have higher scorer uncertainties, but not significantly different in coherency scores.
Additional comparison: Hybrid ToT - Coherency scores not different, but ToT and Hybrid both cheaper than ToT with A* star

CONCLUSION

- Improves performance on reasoning tasks
- Despite higher computational costs, its stronger results and reduced failed attempts make the tradeoff worthwhile

FUTURE WORK

Addressing failing patterns:

- Game of 24: compound expressions
- Creative Writing: inflated scores
- Mini Crosswords: obscure words

REFERENCES

Medium. (2024). Autoregressive generation. All You Need To Know About LLM Text Generation. Retrieved from <https://medium.com/@javaid.nabi/all-you-need-to-know-about-llm-text-generation-03b138e0ed19>

Yao, S., Yu, D., Zhao, J., Shafraan, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). *Tree of Thoughts: Deliberate Problem Solving with Large Language Models*. arXiv. <https://doi.org/10.48550/arXiv.2305.10601>